

The Web of Financial Regulation*

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Abstract

Financial regulation is organized modularly, with individual provisions linked through extensive cross-references. We show that this reference structure has become substantially more complex in both securities and banking regulation. We develop a measure of regulatory complexity and apply it to Titles 12 and 17 of the Code of Federal Regulations, governing banking and U.S. Securities and Exchange Commission (SEC) regulation. We argue that greater regulatory complexity should reduce firm performance for all firms and should have differential effects on valuation, increasing valuation for listed non-bank firms and decreasing valuation for banks. However, we find that securities regulation complexity decreases performance and firm value, while banking regulation complexity increases performance and value.

JEL Classification: G18, G28, G41, K23.

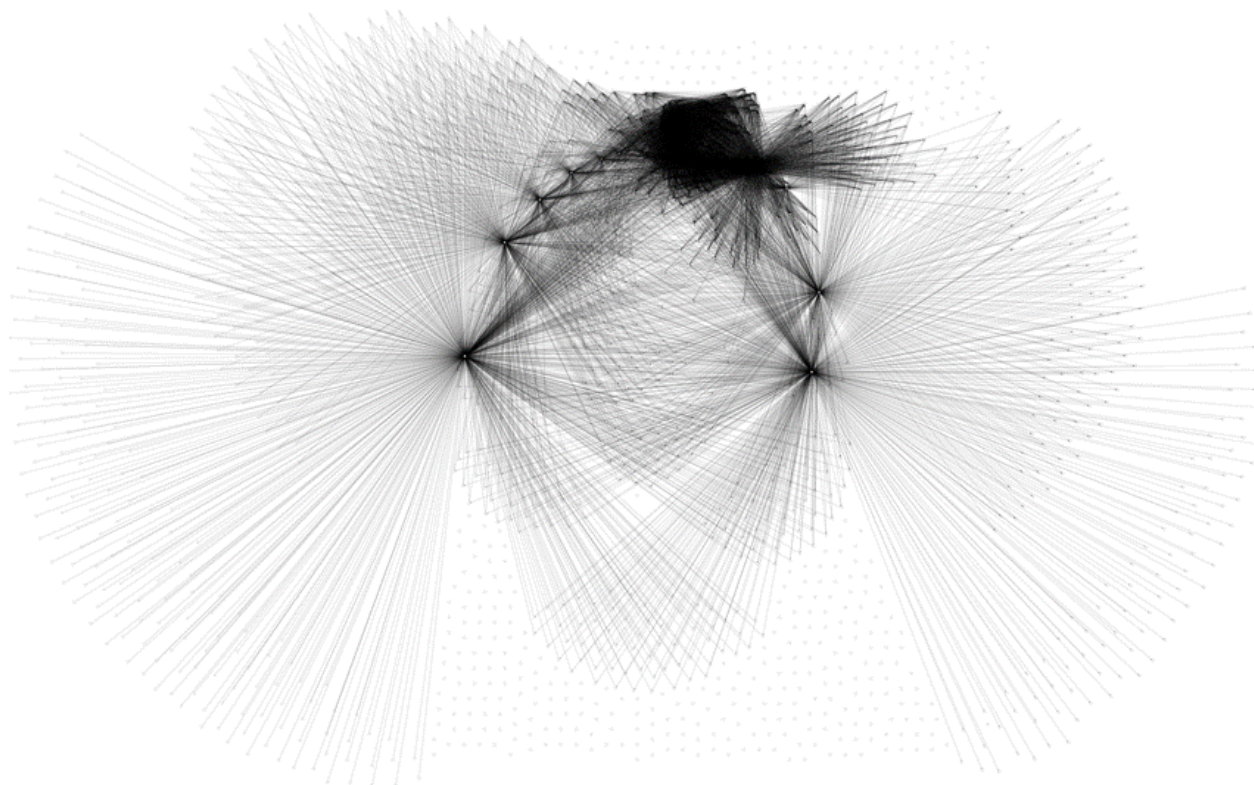
Key words: complexity, regulation, references.

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Figure 1: Illustrating the web of banking regulations. Taking paragraphs in title 12 of the C.F.R. as smallest unit, represented as dots, this links them together through references.



1 Introduction

Financial regulation prescribes actions and therefore must be concise and unambiguous in communicating what is required of regulated entities. While it may be an idealistic aspiration for regulation to read like good prose, the reality is that regulatory texts are dense, technical, and jargon-laden. This jargon is justified only insofar as it is firmly anchored in the regulatory framework through explicit cross-references to the provisions in which terms and concepts are defined. One might envision regulatory texts as having a hierarchical structure, akin to a forest of (inverted) trees. In reality, however, cross-references transform regulatory texts into a dense and intertwined web. As Figure 1 demonstrates for banking regulation in 2024, regulated entities must navigate a highly interconnected network of provisions. This

paper examines how this regulatory web has evolved over recent decades in U.S. securities and banking regulation issued by the Securities and Exchange Commission (SEC) and the Federal Reserve.

From a conceptual perspective, references can be useful in providing structure, particularly when they point to higher-level concepts or general principles. In practice, however, extensive referencing materially increases the difficulty of understanding regulatory requirements. When a provision incorporates other provisions by reference, comprehension of the former is impossible without consulting the latter. This burden escalates sharply when referenced provisions themselves contain further references. Such multi-layered referencing forces readers to traverse long chains of provisions to understand even a single regulatory obligation. As we show in this paper, the average depth of referencing in banking regulation has increased by roughly a factor of three, from about two references in 1980 to approximately six in 2024.

Our analysis is based on the U.S. Code of Federal Regulations (CFR), focusing on Title 12 (“Banking Regulation”) and Title 17 (“Securities Regulation”). We treat paragraphs as the smallest unit of analysis. For each paragraph, we identify references to other paragraphs on an annual basis from 1980 to 2024, thereby constructing the evolving regulatory web for both domains. Figure 1 illustrates this web for banking regulation in 2024.

We argue that internal referencing is a central and previously underappreciated driver of regulatory complexity. To quantify this complexity, we develop a measure based on regulatory workload—the effort required to read and understand regulatory obligations. Existing approaches largely rely on measures of linguistic difficulty, sometimes combined with text length. We extend this literature by explicitly incorporating references. Specifically, we use the Flesch–Kincaid Reading Ease Score (FRES), a standard metric of textual difficulty (see Kincaid et al. (1975) and Flesch (1979)). In the absence of references, we define workload as the product of the FRES score and the number of required “actions” (hereafter operators) imposed on regulated entities, identified using the dictionary of Colliard and Georg (2025). When references are present, we augment workload by adding the workload of referenced provisions, weighted by a fixed, non-negative scaling factor. Regulatory complexity is defined as total workload normalized by the number of operators.

We apply this framework to Titles 12 and 17 of the Code of Federal Regulations, yielding separate measures of complexity for banking and securities regulation. We document a

dramatic increase in regulatory complexity in both domains since 1970.

We then examine the empirical relation between regulatory complexity, operating performance, and market valuation (Tobin’s Q) for non-bank firms and banks. Theory predicts that greater regulatory complexity should reduce firm performance across all firms and increase (decrease) valuation for listed non-bank firms (banks). Both the number of cross-references and our composite complexity measure have substantial explanatory power for these outcomes. In sharp contrast to these predictions, greater complexity in securities regulation is associated with lower performance and firm value, while greater complexity in banking regulation is associated with higher performance and firm value.

This paper contributes to the literature in two ways. First, it documents the evolution of the regulatory web created by internal cross-references and shows that ignoring this structure leads to a systematic understatement of regulatory complexity. Second, prior work studies regulatory complexity along separate dimensions: the scope of regulatory requirements (Al-Ubaydli and McLaughlin (2017); Colliard and Georg (2025)), nesting through references (Li et al. (2008); Katz and Bommarito (2014)), and linguistic difficulty (Kincaid et al. (1975); Flesch (1979)). This paper integrates all three dimensions into a unified measure of regulatory complexity, providing a more complete characterization of the evolution and economic relevance of financial regulation.

The remainder of the paper is structured as follows. The next section reviews the related literature. Section 4 introduces our novel reference-based measure of complexity. The subsequent section examines its empirical properties in the context of banking and securities regulation. The final section 6 concludes.

2 The Literature of Financial Complexity

Regulatory complexity has long been recognized as a defining feature of modern financial regulation. Existing literature distinguishes between different sources of complexity, most notably intrinsic and extrinsic complexity. While intrinsic and extrinsic complexity are conceptually distinct, they are likely intertwined in practice, with each potentially reinforcing the other. Importantly, only extrinsic complexity lends itself to realistic empirical measurement, motivating its focus in applied analysis.

2.1 Intrinsic Complexity

Intrinsic complexity arises from the nature of the regulatory problem itself, such as convoluted policy objectives or the use of sophisticated formulas relying on non-robust input parameters. This strand of complexity has been extensively discussed in the context of banking regulation and systemic risk, with seminal contributions highlighting how complex regulatory environments may facilitate regulatory capture or obscure accountability. The issue itself has received large attention and is often discussed, see, e.g., Rochet (2010), Gai et al. (2011), and Gai et al. (2019).

2.2 Extrinsic Complexity

The previous subsection studied the inherent complexity coming from the issue under consideration. By contrast, extrinsic complexity stems from the way regulation is written and communicated, including unclear drafting, dense language, or provisions that are open to misinterpretation.

Starting with Christainsen and Haveman (1981) and Gray (1987), a substantial literature examines the legal complexity of financial regulation.¹ This literature focuses on the textual formulation of regulatory provisions to assess the effort required to understand and comply with regulation. A variety of “text-based” measures have been developed to capture different dimensions of textual complexity (Li et al. (2008); Colliard and Georg (2025)).

A central challenge is defining complexity in a way that can be quantitatively linked to regulatory costs, which has led to a proliferation of measures. Broadly, these measures fall into three categories, depending on the aspect of complexity considered.

The first category captures lexicographical or readability-based complexity, using indicators such as word length or repetition (Amadxarif et al. (2019); de Lucio and Mora-Sanguinetti (2022)), building on classic readability measures (Kincaid et al. (1975); Flesch (1979)). The second category focuses on regulatory scope, typically measured by text length or the frequency of instruction-related keywords (Li et al. (2008); Al-Ubaydli and McLaughlin (2017); Colliard and Georg (2025)), conceptualizing regulation as a sequence of implementable instructions. The third category examines the reference structure of regulation,

¹Related studies propose measures of regulatory costs; see, e.g., Ahmed et al. (2010), Calomiris et al. (2020), Coffey et al. (2020), Kalmenovitz (2023), and Trebbi et al. (2023).

modeling texts as networks of cross-references and applying graph-theoretic measures such as density (Li et al. (2008); de Lucio and Mora-Sanguinetti (2022)). Dense reference structures may impede comprehension and compliance by increasing cognitive load and information overload (Eppler and Mengis (2004); Chewning and Harrell (1990); Hwang and Lin (1999)).

3 References and its Web

This paper argues that cross-references within regulations are a key component of financial regulation, creating a web that has been ignored largely so far. We focus on titles 12 in the U.S. Code of Federal Regulations (CFR) that covers banking regulation and on title 17 of CFR that covers the Securities and Exchange Commission’s regulation.

Throughout, our analysis takes the paragraph as the basic unit of study, examining both references within a paragraph and those between paragraphs. This choice is motivated by practical considerations: regulations themselves typically use the paragraph as the smallest explicit unit of reference, making it a natural starting point. Conceptually, however, our approach is flexible and could be adapted to coarser or more granular units.²

3.1 An illustrating referencing: An example

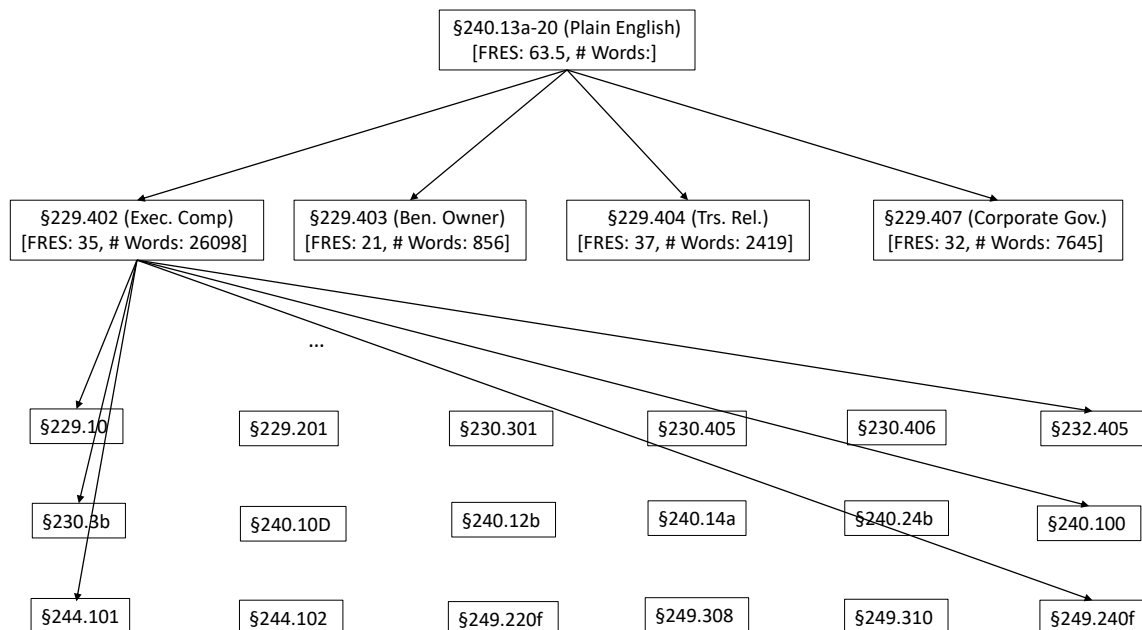
To illustrate referencing we take a look at the paragraph from Securities and Exchange Commission regulation that prescribe plain English to its regulatory texts, 17 CFR § 240.13a-20:

“Any information included or incorporated by reference in a report filed under section 13(a) of the Act (15 U.S.C. 78m(a)) that is required to be disclosed pursuant to Item 402, 403, 404 or 407 of Regulation S-K (§ 229.402, § 229.403, § 229.404 or § 229.407 of this chapter) must be presented in a clear, concise and understandable manner.”

This was incorporated in 1997 into the Securities and Exchange regulation and shortly

²Using coarser units would shift some references from occurring between units to occurring within them, whereas finer units would have the opposite effect. The former adjustment is straightforward, while the latter would require machine reading techniques, which fall beyond the scope of this paper.

Figure 2: Illustrating the Reference Structure of the SEC’s Plain English Regulation. All references are within title 17 of the United States Code of Federal Regulations.



thereafter, the Gramm–Leach–Bliley Act of 1999 extended this requirement to banking regulators³.

Figure 2 provides an illustration of “references” and its structure. Therein, our starting point is the Securities and Exchange Commission’s regulation on plain English, title 17 § 240.13a-20 in the U.S. Code of Federal Regulations (CFR). This paragraph references four other paragraphs within the same title 17 that detail specific issues: §229.402 (entitled “Executive Compensation”), §229.403 (entitled “Security ownership of certain beneficial owners and management”), §229.404 (entitled “Transactions with related persons, promoters and certain control persons”) and §229.407 (entitled “Corporate Governance”). Each of these four paragraphs refers to other paragraphs within title 17; for simplicity Figure 2 exhibits *only* the 18 paragraphs that §229.402 refers to: §229.10, §229.201, §230.301, §230.405, §230.406,

³“Each Federal banking agency shall use plain language in all proposed and final rulemakings published by the agency in the Federal Register after January 1, 2000. (...) For purposes of this section, the term ‘Federal banking agency’ has the meaning given that term in section 3 of the Federal Deposit Insurance Act.” (Excerpt from Section 722 of the Gramm-Leach-Bliley Act).

§232.405, §230.3b, §240.10D, §240.12b, §240.14a, §240.24b, §240.100, §240.101, §240.102, §249.220f, §249.308, §249.310 and §249.240f. We do not depict this here, but each of these 18 paragraphs may reference other paragraphs of title 17.

Figure 2 focuses on a single paragraph (“plain English”). Yet even understanding this paragraph in isolation quickly becomes a tedious task, as it requires consulting at least four other referenced paragraphs, which in turn point to many more. (Here we depict only 18 referenced by one of them.) The process demands working through long stretches of text. Although each is, in principle, written in “plain English,” it is difficult to hold all of them in mind simultaneously. As the notion of “mental overload” suggests, each additional reference compounds the difficulty. This observation underlies our approach of treating references explicitly as a key component of regulatory complexity.

3.2 The Web of Current and Historical Financial Regulation

First, we examine the provisions of the U.S. Securities and Exchange Commission (SEC) from 1970 to 2024, codified in Title 17 of the Code of Federal Regulations (CFR). Second, we analyze capital rules in banking regulation, codified in Title 12 of the CFR. We do so year by year.

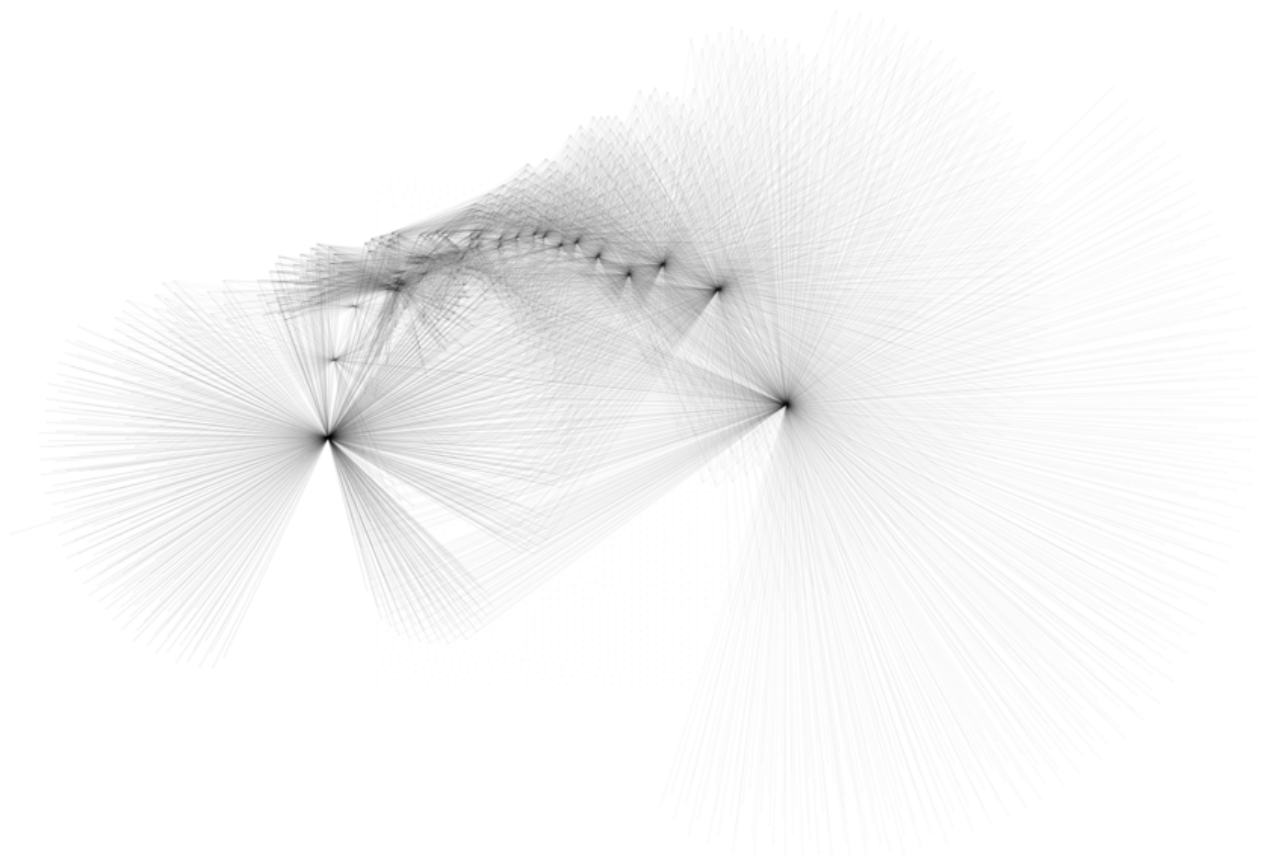
The SEC’s regulations primarily concern listed companies.⁴ This is of interest because rising regulatory burdens have been linked in the literature to both the decline in the number of listed companies (Engel et al. (2007)) and increased market concentration (Singla (2023)) in the United States.

Table 1 provides some descriptive statistics on both financial regulation. For now we are interested only in the average number of references (defined as the total number of references divided by the total number of paragraphs). We note that the average across years is 3.3 for the securities and 4.0 for the banking regulation. Both values appear fairly large with the banking regulation being even larger.

Figure 4 looks at these parameters over the years to provide additional insights. Over all the years, the number of paragraphs increase for both regulations, yet banking regulation has about half the number of paragraphs in all years. Moreover, we note that both regulations started with an average of one reference per paragraph in the 1980’s, yet both finish with

⁴They also address other aspects of creditor protection, which we omit for simplicity.

Figure 3: Illustrating the web of securities regulations. Taking paragraphs in title 17 of the C.F.R. as smallest unit, represented as dots, this links them together through references.



much larger average references per paragraph in current years. Both increased one along the other up to the end of the millennium, finishing then with four references per paragraph, on average, a much higher number than at the beginning of our observation period. In the new millennium, the average number of references in banking regulation increased further considerably (to more than six these days), while the average references in securities regulation largely leveled off.

4 A Measure of Regulatory Complexity

Our literature review noted that the difficulty of legal texts is commonly assessed through readability, potentially coupled with some measure of length, e.g. the number of words or the

Table 1: Descriptive Statistics of Financial Regulation.

	Count	Mean	Std. Dev.	Min	25%	50%	75%	Max
SEC # parag. (thsd.)	47	1.782	0.291	1.292	1.490	1.844	2.067	2.163
SEC # words (mill.)	47	0.914	0.292	0.440	0.638	0.891	1.155	1.396
SEC # syllables (mill.)	47	1.537	0.498	0.729	1.069	1.491	1.942	2.367
SEC FRES	47	32.92	2.248	29.19	30.88	32.96	34.75	38.24
SEC # operators (thsd.)	47	31.70	8.935	16.18	23.31	31.52	39.22	45.97
SEC avg. ref.	47	3.253	1.039	0.980	2.718	3.543	4.066	4.626
SEC complexity (thsd.)	47	0.700	0.430	0.0242	0.217	0.862	1.088	1.281
FED # parag. (thsd.)	42	0.984	0.245	0.711	0.790	0.901	1.159	1.466
FED # words (mill.)	42	0.794	0.288	0.479	0.584	0.664	1.027	1.378
FED # syllables (mill.)	42	1.337	0.498	0.786	0.973	1.113	1.714	2.342
FED FRES	42	39.46	2.309	34.69	38.24	39.69	41.09	43.40
FED # operators (thsd.)	42	27.93	10.78	17.16	20.47	22.55	34.70	49.16
FED avg. ref.	42	4.040	1.621	1.386	2.195	4.435	5.139	6.887
FED complexity (thsd.)	42	0.269	0.180	0.0263	0.151	0.251	0.308	0.671

number of operators. This section argues that the web of references should be considered as an additional component in the difficulty of mastering legal texts and, moreover, introduce a joint measure, called complexity, that reflects all three aspects.

4.1 Referencing adds to Complexity

The empirical literature on regulatory text complexity has traditionally relied on two primary approaches: readability and length. Readability is commonly measured using the Flesch–Kincaid Reading Ease Score, which ranges from 0 to 100, with higher scores indicating greater ease of comprehension. Regulatory standards in the United States have mandated the use of plain English in both securities and banking regulation since the late 1990s, reflecting concerns about accessibility and transparency. Length-based measures, such as word counts, provide a more mechanical proxy for complexity but are limited in their ability to capture cognitive burden. More recent work conceptualizes regulation as a form of computer code, emphasizing the role of legal “operators” such as “shall” or “required,” see Colliard

Figure 4: The number of paragraphs (left-hand) and the average number of references of paragraphs (right-hand) over the years, for banking regulation (Fed) and securities regulation (SEC).

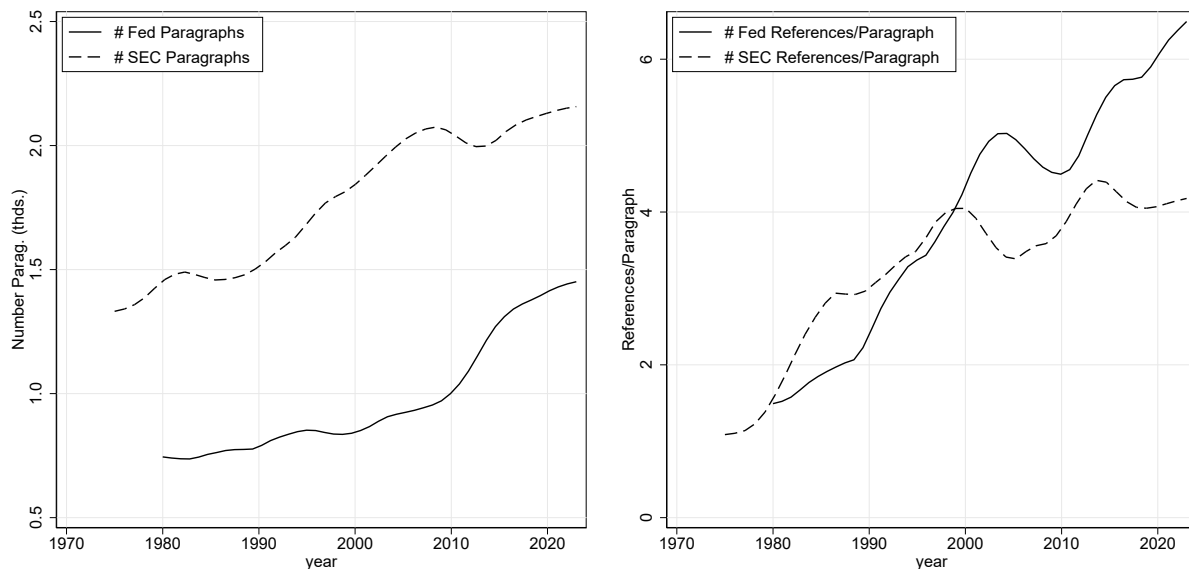
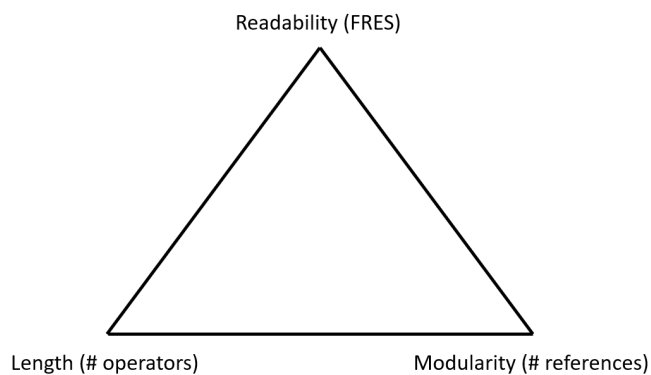


Figure 5: The three dimensions of complexity that we consider.



and Georg (2025).⁵

⁵Although this approach is conceptually appealing, empirical evidence suggests that operator counts are closely correlated with text length and therefore offer limited additional explanatory power, while leaving the economic implications of complexity largely unexplored.

In practice, agencies such as the Federal Reserve and the OCC often include a boilerplate assurance along the lines of: “The Board has sought to present the final rule in a simple, straightforward manner. The Board did not receive any comment on its use of plain language.” While such statements appear to suggest that financial regulation is not unduly complex, this sits uneasily with the longstanding critique that the regulatory framework remains excessively complicated.

What, then, are we missing when we talk about complexity? The excerpt above points directly to the central issue of this paper. To begin, consider the Flesch–Kincaid Reading Ease Score (henceforth FRES), a widely used metric for evaluating plain English writing, see Kincaid et al. (1975) and Flesch (1979). The quoted passage receives a FRES of 63.5, suggesting that a reader with an eighth- or ninth-grade education should be able to follow it with ease. By this measure, the text qualifies as plain English.

Yet comprehension is not merely a function of sentence length or word choice. Full understanding is complicated by the dense web of cross-references to other regulatory provisions.

Modern regulation increasingly proceeds through higher-level principles that are referenced across documents, creating a dense web of cross-references. Such modularity can reduce overall length and lower cognitive load by allowing readers to process information in structured chunks, consistent with established findings in cognitive psychology on information processing limits. However, excessive reliance on references may overwhelm readers, particularly when they must simultaneously integrate multiple distinct provisions. Insights from the psychological and learning literature on split attention suggest that juggling several separate sources of information can impair comprehension and focus. Accordingly, while modularity may enhance regulatory efficiency up to a point, excessive referencing may itself become a source of extrinsic complexity with meaningful economic consequences, see, e.g. Miller (1956) and the “split attention” literature.

Our paper focuses on measuring precisely this added dimension of complexity, the complexity generated by the structure of regulatory referencing. Overall then, we consider three dimensions of complexity, as illustrated in Figure 5: (1) readability (the Flesch-Kincaid reading ease score); (2) length (the number of operators); (3) modularity (the number of references per paragraph)

4.2 Workload of a Paragraph without References

The Flesch-Kincaid Reading Ease Score (FRES) of a text (or of any paragraph of it), denoted L , was introduced by Kincaid et al. (1975) and Flesch (1979). It is a widely used measure of the readability of texts. Higher FRES values correspond to easier texts, with scores above the conventional cutoff of 60 typically classified as “plain English.”

While the FRES does not account for references, we adopt it as the foundation of our approach. For our purposes, we require a measure F of reading complexity, such that higher values indicate greater complexity. We therefore define F by a simple linear transformation of the FRES L :

$$F = 2 - \frac{L}{60}.$$

Under this definition, higher F values correspond to more complex texts, and the “plain English” cutoff aligns with $F = 1$.

By construction, the FRES is unbounded, and values outside the interval $[0, 1]$ are possible. In fact, the literature documents examples with FRES values in the range of -500, which would translate into complexity levels around $F \approx 6$ under our transformation.

The FRES, and therefore our complexity measure, rests on two components: the average number of words per sentence and the average number of syllables per word. Put differently, text complexity is determined by the (average) length of sentences and the (average) length of words.

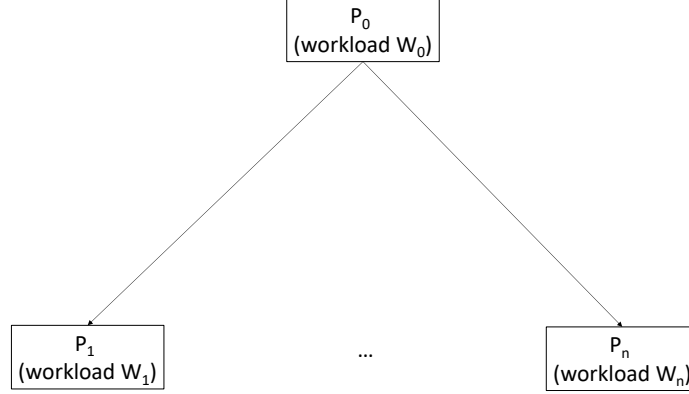
However, legal texts such as laws, regulations, and orders introduce an additional source of complexity. They prescribe actions using specialized vocabulary: terms such as “(dis-)allow”, “deem”, “exempt”, “mitigate” and related nouns as well as adverbs. Such words, though rare in everyday prose, play a central role in regulatory language and make interpretation more demanding. In short, they contribute to complexity and should be reflected in a suitable measure.

We build on Colliard and Georg (2025), who cataloged these specialized terms, referred to as operators. Absent references, a straightforward approach is to count the number of operators N_A of a text “A” and define its “workload” W_A as the product of this count with the complexity score:

$$W_A^N = F_A \cdot N_A \text{ and } C_A = W_A/N_A, \tag{1}$$

where C_A represents the relative (or proportional) workload, i.e., the average complexity per

Figure 6: Illustrating the Reference Structure of a Paragraph.



regulatory operator.⁶

In this way, $C_A = F_A$ corresponds to the baseline case: the average complexity per operator in a text without references.

4.3 Workload of a Paragraph with References

We now extend our complexity measure to account for the case of a paragraph that references other paragraphs.

Without loss of generality, let paragraph P_0 reference n other paragraphs, denoted P_i , $i = 1, \dots, n$. Let F_0 and N_0 denote the reading complexity and the number of operators of P_0 , respectively. For each referenced paragraph P_i , we denote its workload by W_i . Figure 6 provides an illustration.

Reading paragraph P_0 requires not only processing its own content but also consulting all referenced paragraphs P_i ($i = 1, \dots, n$), each with workload W_i . We therefore define the total workload of P_0 , incorporating its references, as

$$W_0^R = W_{P_0 \text{ with references}} = F_0 N_0 + c \sum_{i=1}^n W_i, \quad (2)$$

⁶Alternative specifications are possible. For instance, one could assign fractional weight to operators, e.g. $W_A = F_A \cdot (1 + N_A/10)$.

where $c > 0$ is a fixed parameter that controls how much a referenced paragraph contributes to the workload of the paragraph that “origin”.

A “naïve” way to assess the complexity in the situation depicted in Figure 6 is to consider the workload of paragraph P_0 in isolation, disregarding its reference structure.⁷ Based on the definition in equation (1), this approach would yield

$$W_0^N = W_{P_0 \text{ without references}} = F_0 N_0. \quad (3)$$

If the referenced paragraphs P_i , $i = 1, \dots, n$, do not themselves reference further paragraphs, then their workloads are given directly by

$$W_i = F_i \cdot N_i,$$

where F_i and N_i denote the reading complexity and the number of operators of paragraph $i = 1, \dots, n$, respectively.

If, however, the referenced paragraphs contain further references, the scheme in equation (2) can be applied, provided that the workload W_i of each paragraph is well-defined. In this way, equation (2) offers a recursive procedure to determine total workload, as long as every chain of references eventually terminates in paragraphs for which workload can be computed as in subsection 4.2.

This assumption, however, does not always hold. A classic difficulty, known as the dictionary paradox, arises when definitions (or references) form a closed loop: a word in a dictionary refers to another, which refers to another, until eventually circling back to the original word. Analogously, regulatory references may create loops among paragraphs, preventing the workload of any paragraph in the cycle from being defined under the recursive scheme.

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⁷This should not be misunderstood as ignoring the referenced paragraphs entirely. They still contribute when determining the workload of the text as a whole; see the discussion in the next subsection.

scheme.

4.4 Workload and Complexity of the Entire Text

The first step is to identify in a regulatory text the paragraphs $P_i, i = 1, \dots, m$ and all the references. We discussed on Figure 2 the task of identifying references on the (part of) paragraph §240.13a describing “plain English” and stress that this has to be done for all paragraphs in the regulatory text analogously.

The next step is then to note that the previous two subsections defined the workload W_i^R for every paragraph i therein. We then define the workload $W_{1,\dots,m}^R$ of the entire text as the sum of all these workloads.

Finally, using the number of operators N_i in each paragraph, we define the complexity of the entire text as

$$C_{1,\dots,m}^R = \frac{W_{1,\dots,m}^R}{\sum_{i=1}^m N_i},$$

One may view this as an average Flesch-Kincaid complexity of paragraphs, weighted by the referencing in the text.

4.5 Redoing the Workload in Assessing Regulatory Paragraphs

Disregarding the complications introduced by loops, the recursive structure in equation (2) implies that the “naïve workload” of a paragraph may be counted multiple times. To illustrate, consider the case shown in Figure 6, assuming that the referenced paragraphs do not themselves contain further references. In this case, the workload of a referenced paragraph P_i is simply $F_i N_i$.

Under the naïve approach (ignoring references), the workload of paragraph P_0 is

$$W_0^N = F_0 N_0.$$

By contrast, under the recursive approach that accounts for references, it becomes

$$W_0^N = F_0 N_0 + c \sum_{i=1}^n F_i N_i.$$

Consequently, the workload of the entire text is given by

$$W_{\text{text}}^N = F_0 N_0 + \sum_{i=1}^n F_i N_i$$

under the naïve approach, and by

$$W_{\text{text}}^R = W_0^R + \sum_{i=1}^n F_i N_i = F_0 N_0 + (c + 1) \sum_{i=1}^n F_i N_i$$

with our recursive approach, the one that takes into account references.

The key difference lies in the multiplicative factor: the naïve approach counts the contribution of the referenced paragraphs only once, whereas our recursive approach counts it $c + 1 > 1$ times. This reflects the core insight of our framework: referencing generates additional workload because each referenced paragraph must be reassessed in full whenever it is encountered.⁸

4.6 Complexity of Financial Regulation Over the Years

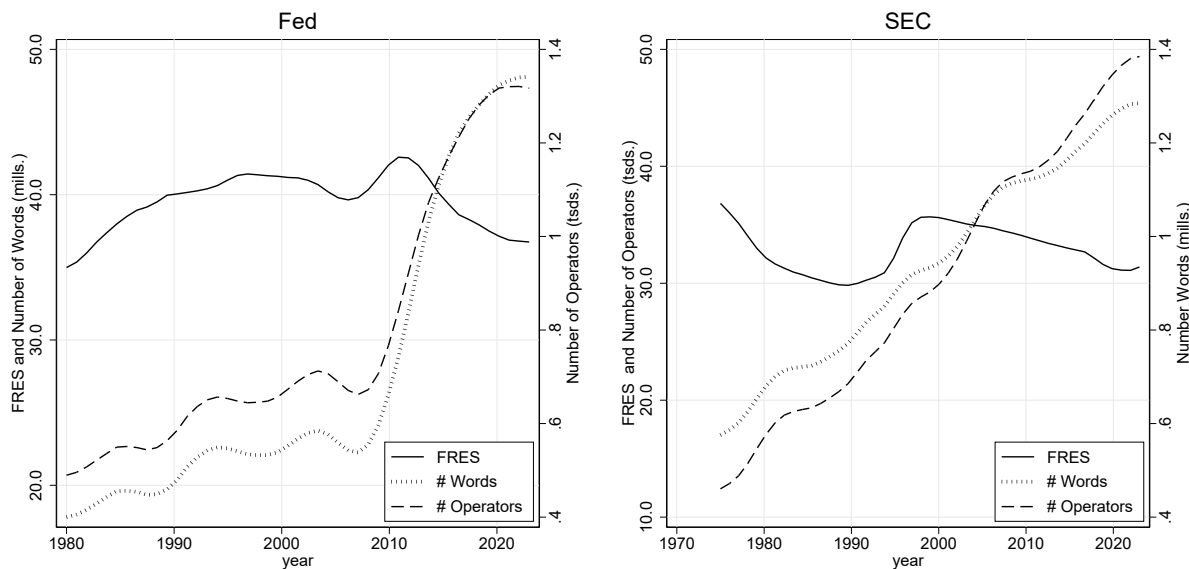
Figure 4 showed right-hand the average references (in paragraphs) over the years, for banking and securities regulation. We then argued that our complexity incorporates two additional dimensions: readability and length, see Figure 5. This subsection first illustrates how both evolved over the years and, finally, illustrates our complexity measure, over the years.

Figure 7 illustrates how readability/length in regulation evolved from 1970 to 2024 according to various basic measures from the literature. Regarding securities regulation over that time period, the number of words roughly tripled, the number of operators increased almost four-fold and all these increased almost linearly over time.⁹ Relatively, over these years, the number of words/operators in banking regulation similarly. However the increase is not linearly over the entire period, rather they increase only slightly up to about 2010 and

⁸Of course, deep familiarity with a regulation may allow specialists to navigate such references without re-reading them from scratch. Nevertheless, our framework captures the workload faced by a non-specialist reader.

⁹The number of operators is the measure put forward by Colliard and Georg (2025). Our analysis here recommends using it, since it increased considerably over time, more than a simple counting of words would suggest.

Figure 7: Complexity of Banking (Fed) and Securities (SEC) Regulation over the years according to basic measures of it.

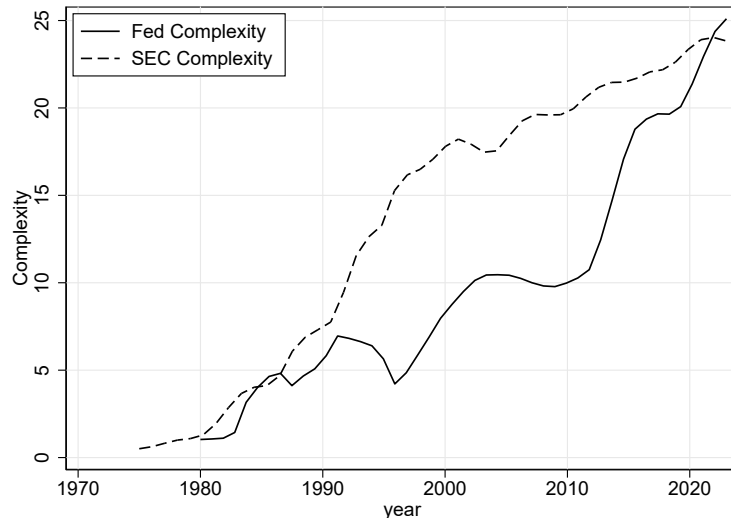


increases much stronger since then.

Moreover, both plots illustrate the Flesch-Kincaid reading easy score (FRES). We see that the linguistic complexity of both regulations is always at college level, with banking regulation having (slightly) higher readings than securities regulation, i.e. the former is somewhat easier to read. (We recall that lower FRES readings correspond to more complicated texts.)

Figure 8 illustrates the relative increases in our complexity measure, setting the complexity measure equal to 1 in 1980 for both securities and banking regulation. For both regulations, complexity increased roughly 8-fold up to the early 1990's. Up to today, complexity for both regulations increased 3-fold, overall representing a 25-fold increase since 1980. However, between the 1990's and today, complexity diverges. Since then, the increase differs from banking to securities regulation: the increase in the complexity of securities regulation occurred mostly before the millennium, whereas the increase in banking regulation occurred mostly since 2010.

Figure 8: Complexity of Banking (Fed) and Securities (SEC) Regulation over the years according to our measure. For better comparability we rescaled both measures to be equal to 1 in 1980.



5 How References and Complexity Impact Firm Performance and Values

This section studies the financial impact of what we studied so far. Specifically, we now study how complexity, assessed through our novel measure, affects the performance and valuation of firms. Therein, we study both (non-bank) firms and banks and look at our two major areas of U.S. financial regulation, banking and securities regulation.

5.1 Hypotheses

Public regulation is generally justified by the existence of market failures, such as market power, externalities, or asymmetric information between buyers and sellers, see, e.g. chapter 9 in Freixas and Rochet (2008). In the financial sector, regulation mainly takes the form of securities regulation and banking regulation, each with distinct objectives.

Securities regulation aims to reduce market power and mitigate information asymmetries in financial markets. Although compliance with such regulation is costly, it is expected

to improve transparency and investor protection, thereby increasing the value of securities. Banking regulation, in contrast, focuses on protecting depositors and the government, particularly through deposit insurance mechanisms. By promoting the soundness and stability of banks, these regulations protect debt holders but may reduce value for equity holders due to increased constraints and costs.

Regulatory complexity plays a crucial role in shaping these outcomes. Greater complexity increases operating costs, as financial institutions must devote more resources to compliance, specialized personnel, and external consulting.

Hypothesis 1 (Performance Hypothesis) *Higher regulatory complexity is associated with higher operating costs for regulated institutions.*

If regulatory complexity reflects the underlying economic complexity of financial activities—that is, if regulation is endogenous and set at an optimal level—then its impact on firm value should align with regulatory objectives.

Hypothesis 2 (Value Hypothesis – Securities Regulation) *Greater complexity in securities regulation is value-improving for firms subject to such regulation.*

Hypothesis 3 (Value Hypothesis – Banking Regulation) *Greater complexity in banking regulation is value-decreasing for equity holders, reflecting the trade-off between financial stability and shareholder value.*

5.2 Data and Descriptive Statistics

As the data basis for non-banking firms we use a CRSP/Compustat dataset covering the years 1986 to 2023, accessed through WRDS. We restrict ourselves to companies that are not banks, are headquartered in the United States and listed on one of the major exchanges (NYSE, American SE, NASDAQ).

In the United States, Bank Holding Companies (BHCs) have been required to report their salary expenses within the FR Y-9C reports, which they must submit quarterly to the Fed, making these data publicly available. This provides a granular dataset that we use for our analysis of banks.

Table 2: Descriptive Statistics of Financial Characteristics.

	Count	Mean	Std. Dev.	Min	25%	50%	75%	Max
Panel A: Non-Banks								
ROA (%)	141962	2.328	156.8	-115.1	-0.951	5.914	12.08	27061.9
ROE (%)	136883	10.88	5908.5	-263.7	-1.368	15.14	29.57	690479.6
Tobin Q	160261	2.433	42.21	0.0210	1.042	1.390	2.202	12253.4
log(Assets)	161328	1.430	2.316	-11.26	-0.238	1.317	2.977	10.67
Panel B: Banks								
ROA (%)	9274	2.276	1.840	-1.765	1.703	2.215	2.778	77.55
ROE (%)	9267	29.42	412.5	-23.63	17.22	23.58	31.19	39136.4
Tobin Q	11815	1.052	0.172	0.0834	1.002	1.036	1.079	8.314
log(Assets)	11829	3.221	1.685	-0.915	2.018	2.846	4.091	10.16

Throughout, we control for firm size through the inflation-adjusted (logarithm of) total assets. We measure performance through the return on assets (ROA), defined in the usual way and valuation through Tobin Q. Moreover, we consider the logarithm of Tobin Q, since its distribution is highly skewed.

5.3 Listed (Non-Bank) Firms

Table 3 studies in four regressions how complexity of securities regulation affects (non-bank) firm performance. The first regression studies how the three dimensions affect performance, whereas the other three regressions study the impact of our complexity measure on it.

We find that the increasing the average references or the length of regulation or making the text more readable, decreases firm performance. Moreover, increasing the complexity of securities regulation decreases performance. Overall, this confirms hypothesis 1 for (non-bank) firms.

Table 4 studies in five regressions how complexity of securities regulation affects the logarithm of (non-bank) Tobin Q. The first regression studies how the three dimensions affect performance, whereas the other four regressions study the impact of our complexity

Table 3: Studying how complexity and its three dimensions affect (non-bank) firm performance. Coefficient statistics are calculated using robust methods; t -statistics are reported in parentheses; significance levels refer to * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1) ROA	(2) ROA	(3) ROA	(4) ROA
SEC avg. ref.	-3.242*** (-2.59)			
SEC FRES	-0.926*** (-3.81)			
SEC # operators (thsd.)	-0.736* (-1.65)			
SEC complexity (thsd.)		-27.30*** (-2.81)	-24.99*** (-5.26)	-23.65** (-2.14)
SEC complexity (thsd.) $\times$ log(Assets)				-2.817* (-1.77)
log(Assets)	14.42*** (3.29)	14.47*** (3.52)	14.58*** (3.28)	17.12*** (4.88)
Year			-0.0825 (-0.28)	
Constant	40.68*** (2.72)	-3.720 (-1.32)	159.3 (0.27)	-6.824* (-1.70)
Obs.	146721	146721	146721	146721
R^2	0.337	0.337	0.337	0.337
F	17.56	10.81	21.95	12.41

measure on it.

We find that the increasing the average references or the length of regulation or making the text less readable, decreases firm Tobin Q. In line with this, the regression in column (2) shows that increasing the complexity of securities regulation decreases Tobin Q. The following three regressions delve deeper into this, looking at interaction terms with years (column 3) and size (columns 4, 5). Regression (3) shows that (roughly) after the year $46.31/0.0233=1987$ increasing the complexity of securities regulation decreased Tobin Q. The remaining two regressions (4, 5) study the interaction of complexity with size and show that securities complexity increases Tobin Q for larger firms. Overall, our analysis here

Table 4: Studying how complexity and its three dimensions affect the logairthm of (non-bank) Tobin Q. Coefficient statistics are calculated using robust methods; t -statistics are reported in parentheses; significance levels refer to * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1) Tobin Q	(2) Tobin Q	(3) Tobin Q	(4) Tobin Q	(5) Tobin Q
SEC avg. ref.	-0.00940*** (-3.62)				
SEC FRES	0.00398*** (4.82)				
SEC # operators	-0.00729*** (-19.30)				
SEC complexity		-0.158*** (-19.65)	46.31*** (18.80)	0.100*** (6.90)	0.0298** (2.43)
SEC complexity $\times$ Year			-0.0233*** (-18.76)		
Year			0.0281*** (17.26)	-0.00273*** (-6.12)	
log(Assets)			-0.135*** (-39.43)	-0.178*** (-39.55)	-0.177*** (-39.04)
SEC complexity $\times$ log(Assets)				0.0493*** (13.57)	0.0445*** (12.68)
Constant	0.628*** (22.54)	0.615*** (82.88)	-55.41*** (-17.08)	6.042*** (6.85)	0.648*** (66.59)
Obs.	146002	146002	146002	146002	146002
R^2	0.617	0.617	0.633	0.633	0.632
F	156.77	386.15	652.52	598.65	707.74

rejects hypothesis 2 for (non-bank) firms.

Overall, our analysis of securities regulation for (non-bank) firms shows that increasing regulatory complexity decreases performance and Tobin Q, confirming hypothesis 1 and rejecting hypothesis 2.

5.4 Banks

Whereas the previous subsection studied the complexity of securities regulation, we now take a look at the complexity of banking regulation. Therein, we also take a look at listed banks, which are affected by the complexities of both regulations.

Table 5 presents the performance regression results. Panel A studied only listed banks and therein uses Compustat data to calculate ROA. Moreover, here we study the impact of both banking and securities regulation.¹⁰ Panel B looks at all bank holding companies and uses the Federal Reserve’s database (based on the FRY-9C reports) to calculate ROA. Here we study only the impact of banking regulation. Other than that, Table 5 is organized similar to Table 3. The first regression in each Panel studies how the three dimensions of complexity (average references, readability/FRES, number of operators) affect performance, whereas the other two/four regressions in each Panel study the impact of our complexity measure on it.

We see that the results on securities regulation are roughly in line with those in the previous subsection about (non-bank) firms. Lower FRES, i.e. more complex texts in terms of wording decrease ROA. Increasing the number of operators, the measure of complexity put forward by Colliard and Georg (2025) decreases performance. Moreover, increasing the average references or our measure of complexity both decrease performance.¹¹ Overall, this confirms hypothesis 1 regarding the impact of *securities regulation* on listed banks.

The picture is different when we look at banking regulation. The regressions in columns (1, 4) show that increasing the average reference, decreasing readability or increasing the number of operators *increases* performance. Similarly, increasing the complexity of banking regulation increases performance. Overall, this rejects hypothesis 1 regarding the impact of *banking regulation* on banks.

Finally, we take a look at the impact of regulation on the valuation of listed banks: Table 6 how it drives the logarithm of Tobin Q. (Here, we focus on listed banks, as the market value is observed only for listed ones.) Analogous to our analysis of performance for listed

¹⁰The SEC section of Title 17 CFR imposes regulatory requirements both on all companies whose shares are publicly traded (on exchanges), including financial institutions, especially banks. Hence, listed banks should be impacted by both financial regulations and we now study these jointly. For this, we study measures of securities complexity and of banking complexity.

¹¹The regression in column (2) suggests that this turns around for larger banks. This may also explain the sign in the regression in column (3).

Table 5: Studying how complexity and its three dimensions affect bank performance. Coefficient statistics are calculated using robust methods; t -statistics are reported in parentheses; significance levels refer to * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Panel A: Listed Banks			Panel B: All Banks				
	(1) ROA	(2) ROA	(3) ROA	(4) ROA	(5) ROA	(6) ROA	(7) ROA	(8) ROA
FED avg. ref.	0.146*** (3.73)			0.0725*** (5.46)				
FED FRES	-0.0598*** (-5.95)			-0.0488*** (-8.24)				
FED # operators	0.0356*** (13.82)			-0.00264* (-1.80)				
FED complexity		1.202 (1.14)	3.670*** (7.46)		0.345*** (5.04)	1.187*** (5.24)	0.345*** (5.04)	2.232*** (10.17)
FED complexity $\times$ log(Assets)		-0.0515 (-0.26)				0.0497 (1.16)		
SEC avg. ref.	-0.0466** (-2.00)							
SEC FRES	0.116*** (14.41)							
SEC # operators	-0.101*** (-15.37)							
SEC complexity		-1.313*** (-2.72)	1.964*** (12.04)					
SEC complexity $\times$ log(Assets)		0.296*** (2.90)						
log(Assets)		-0.470*** (-6.00)	0.0203 (0.43)			-0.324*** (-9.26)		
Year			-0.105*** (-12.94)					-0.0370*** (-10.08)
Constant	1.436*** (3.13)	2.370*** (8.86)	208.1*** (13.00)	2.516*** (9.37)	0.694*** (34.86)	1.537*** (15.73)	0.694*** (34.86)	74.31*** (10.18)
Obs.	11034	11034	11034	13275	13275	9907	13275	13275
R^2	0.331	0.323	0.327	0.319	0.315	0.476	0.315	0.323
F	71.25	16.59	69.60	68.07	25.44	36.61	25.44	52.41

banks, here we study both the impact of securities and of banking regulation. Columns (1, 2) therein study the three dimensions, while the remaining 5 columns look at our complexity measure.

First, we look at securities regulation. Overall, we see that securities complexity decreases

Table 6: Studying how complexity and its three dimensions affect the logarithm of bank TobinQ. Coefficient statistics are calculated using robust methods; t -statistics are reported in parentheses; significance levels refer to * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1) Tobin Q	(2) Tobin Q	(3) Tobin Q	(4) Tobin Q	(5) Tobin Q	(6) Tobin Q	(7) Tobin Q
FED avg. ref.	0.0225*** (10.07)	0.00859*** (4.38)					
FED FRES	-0.00264*** (-5.32)	-0.00502*** (-12.69)					
FED # operators	-0.00203*** (-16.80)	0.00282*** (15.07)					
FED complexity			-8.587*** (-3.62)	0.253*** (13.77)	0.214*** (17.92)	0.381*** (15.52)	-47.46*** (-17.81)
FED compl. $\times$ log(Assets)				-0.0372*** (-9.53)		-0.0471*** (-8.44)	
FED compl. $\times$ Year			0.00432*** (3.67)				0.0237*** (17.92)
SEC avg. ref.		-0.0193*** (-8.97)					
SEC FRES		0.0217*** (21.61)					
SEC # operators		-0.00619*** (-26.71)					
SEC complexity					0.220*** (22.95)	0.178*** (8.58)	53.10*** (23.83)
SEC compl. $\times$ log(Assets)						0.00772* (1.80)	
SEC compl. $\times$ Year							-0.0266*** (-23.87)
Year			-0.00484*** (-13.63)	-0.00385*** (-16.13)	-0.00925*** (-27.97)	-0.00908*** (-22.83)	0.0156*** (16.85)
log(Assets)			0.0163*** (5.06)	0.0258*** (11.16)	0.0116*** (5.87)	0.0185*** (4.39)	0.00660** (2.31)
Constant	0.104*** (3.86)	-0.311*** (-7.44)	9.679*** (13.64)	7.640*** (16.17)	18.29*** (28.04)	18.95*** (24.18)	-31.09*** (-16.84)
Obs.	11024	11024	11024	11024	11024	11024	11024
R^2	0.607	0.660	0.598	0.601	0.618	0.807	0.657
F	182.77	331.08	60.06	92.47	204.53	108.43	280.85

Tobin Q, particularly since the year $53.10/0.0266=1966$. We note some evidence for an increase for larger firms. Overall, we conclude here, in line with our earlier analysis in the

previous subsection, that higher complexity of securities regulation decreases Tobin Q. Thus we reject hypothesis 2, for (the securities regulation of) listed banks.

Finally, we take a look at banking regulation. We see in columns (1, 2) of Table 6 that increasing the average references, decreasing readability or increasing the number of operators all increase Tobin Q. (The impact of the number of operators is negative in column 1, yet we do not control for securities regulation here.) This goes against hypothesis 3, for (the banking regulation of) listed banks.

Looking at columns (3-7) confirms this picture. Regressions (4, 5, 6) show that the complexity of banking regulation increase Tobin Q, albeit less for larger banks (columns 4, 6). Column (7) shows that the complexity of banking regulation decreased Tobin Q up to the year 47.46/0.0237=2002, but increases it afterwards. (The impact of the complexity is negative in column 3, yet we do not control for securities regulation here.) We therefore reject hypothesis 3.

Overall, our analysis of banks shows, in line with our results in the previous subsection, that increasing the complexity of *securities regulation* decreases performance and Tobin Q, confirming hypothesis 1 and rejecting hypothesis 2. Moreover, our analysis of banks shows that increasing the complexity of *banking regulation* increases performance and Tobin Q, rejecting hypothesis 1 and rejecting hypothesis 3.

6 Conclusion

This paper examines the web of cross-references in financial regulation and argues that such cross-referencing is a key and previously underappreciated source of regulatory complexity. We introduce a novel measure of complexity that integrates regulatory length, the use of regulatory operators, and the structure of cross-references into a single metric. In contrast to existing approaches that focus on individual dimensions — such as readability (e.g., Flesch-Kincaid scores) or textual length (e.g., word counts or operator counts; Colliard and Georg (2025)) — our measure captures the joint contribution of these elements and their interactions.

We illustrate the application of this measure using Titles 12 and 17 of the Code of Federal Regulations (CFR), which govern banking regulation and U.S. Securities and Exchange Commission (SEC) regulation, respectively. Relying on the dictionary developed by Colliard

and Georg (2025), we identify regulatory operators and cross-references at the paragraph level. We then compute paragraph-level complexity and aggregate these values to obtain overall measures of complexity for banking and securities regulation.

Our analysis documents a substantial increase in regulatory complexity since the 1970s. In banking regulation (securities regulation), the average number of cross-references per paragraph rose from approximately one in the 1970's to six (four) nowadays. Over the same period, word counts tripled and the number of regulatory operators increased fivefold. Taken together, these changes imply that the overall complexity of both banking and securities regulation is roughly twenty-five times higher today than in 1980.

Finally, we study the economic consequences of regulatory complexity for firm performance (ROA) and firm value (Tobin's Q), examining both banks and non-bank firms. We find that greater complexity in securities regulation is associated with lower performance and firm value, whereas greater complexity in banking regulation is associated with higher performance and value. These findings run counter to much of the existing literature. One possible interpretation is that securities regulation has become excessively complex in ways that decrease market values, while banking regulation may have been shaped, at least in part, by regulatory capture, allowing regulated banks to benefit from increased complexity.

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